

## GALLIVARE LAPLAND, Sweden

WMO: 020490

Lat: 67.1407N

Lon: 20.6413E

Elev: 359

StdP: 97.09

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-31.1 | (c)<br>-28.0 | (d)<br>-34.7                  | (e)<br>0.1 | (f)<br>-31.2 | (g)<br>-31.1 | (h)<br>0.2 | (i)<br>-28.0 | (j)<br>9.0            | (k)<br>-1.6 | (l)<br>7.6 | (m)<br>-3.2 | (n)<br>1.2            | (o)<br>310 | (p)<br>0.712 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.7                    | 24.2            | 15.3 | 22.2 | 14.6 | 20.2 | 13.8 | 17.2                | 22.0 | 15.9 | 20.1 | 14.7 | 18.5 | 2.9                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 15.3                            | 11.3 | 18.8 | 14.1 | 10.5 | 17.5 | 13.0 | 9.8 | 16.2 | 49.7          | 21.9 | 45.8 | 20.1 | 42.4 | 18.4 | 21.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.9               | 6.6  | 5.6 | -35.1                      | 26.9 | 3.3                | 2.4 | -37.5                                              | 28.7 | -39.4      | 30.1 | -41.2      | 31.4 | -43.7      | 33.2 |
|                   |      |     | -35.2                      | 18.7 | 3.3                | 1.4 | -37.6                                              | 19.7 | -39.5      | 20.5 | -41.3      | 21.3 | -43.7      | 22.3 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | -0.2   | -12.9 | -12.4 | -7.5  | -1.1  | 5.2   | 10.7  | 14.1  | 11.7  | 6.8   | -0.2  | -7.5  | -10.6 |  |  |
|                                                                        | DBStd       | 10.80  | 7.43  | 7.68  | 5.90  | 4.54  | 3.98  | 3.64  | 3.09  | 3.23  | 3.47  | 4.49  | 6.03  | 6.99  |  |  |
|                                                                        | HDD10.0     | 4014   | 711   | 626   | 542   | 334   | 158   | 34    | 3     | 19    | 107   | 316   | 525   | 638   |  |  |
|                                                                        | HDD18.3     | 6786   | 969   | 859   | 801   | 584   | 407   | 229   | 136   | 206   | 347   | 574   | 775   | 897   |  |  |
|                                                                        | CDD10.0     | 276    | 0     | 0     | 0     | 0     | 9     | 55    | 131   | 72    | 10    | 0     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 7      | 0     | 0     | 0     | 0     | 0     | 1     | 4     | 1     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 96     | 0     | 0     | 0     | 0     | 4     | 22    | 58    | 12    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 9      | 0     | 0     | 0     | 0     | 0     | 3     | 7     | 0     | 0     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.3    | 1.9   | 2.2   | 2.8   | 2.7   | 2.8   | 2.8   | 2.4   | 2.2   | 2.4   | 2.2   | 1.8   | 2.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 591    | 38    | 33    | 28    | 28    | 42    | 64    | 94    | 77    | 56    | 45    | 47    | 42    |  |  |
|                                                                        | PrecMax     | 733    | 71    | 70    | 68    | 53    | 76    | 111   | 150   | 138   | 135   | 100   | 106   | 132   |  |  |
|                                                                        | PrecMin     | 410    | 3     | 7     | 6     | 2     | 2     | 7     | 41    | 27    | 16    | 5     | 7     | 8     |  |  |
|                                                                        | PrecStd     | 94     | 16    | 17    | 14    | 14    | 20    | 27    | 27    | 31    | 31    | 24    | 24    | 25    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 4.4   | 5.1   | 7.1   | 14.0  | 23.2  | 26.2  | 27.6  | 24.9  | 18.8  | 11.1  | 6.0   | 4.8   |  |  |
|                                                                        |             | MCWB   | 2.1   | 2.6   | 3.1   | 7.7   | 12.3  | 15.2  | 18.1  | 16.9  | 13.1  | 8.3   | 4.5   | 2.5   |  |  |
|                                                                        | 2%          | DB     | 1.9   | 2.7   | 4.6   | 10.0  | 19.1  | 22.8  | 25.0  | 22.7  | 16.2  | 9.2   | 3.9   | 2.2   |  |  |
|                                                                        |             | MCWB   | 0.3   | 0.8   | 1.7   | 4.6   | 11.1  | 13.2  | 16.3  | 15.5  | 11.6  | 7.3   | 2.9   | 0.8   |  |  |
|                                                                        | 5%          | DB     | -0.6  | 0.2   | 3.0   | 8.0   | 15.7  | 20.5  | 23.0  | 20.7  | 14.3  | 7.8   | 1.8   | 0.6   |  |  |
|                                                                        |             | MCWB   | -1.5  | -1.1  | 0.7   | 3.6   | 9.4   | 12.8  | 15.2  | 14.9  | 10.8  | 6.1   | 1.0   | -0.2  |  |  |
|                                                                        | 10%         | DB     | -2.8  | -1.4  | 1.2   | 6.1   | 12.6  | 18.0  | 20.9  | 18.2  | 12.9  | 6.1   | 0.3   | -0.9  |  |  |
|                                                                        |             | MCWB   | -3.4  | -2.2  | -0.8  | 2.6   | 7.2   | 11.5  | 14.5  | 13.5  | 10.0  | 4.5   | -0.1  | -1.5  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 2.5   | 2.9   | 3.3   | 7.5   | 13.6  | 17.2  | 19.2  | 18.5  | 14.1  | 9.1   | 4.6   | 2.6   |  |  |
|                                                                        |             | MCDB   | 4.1   | 4.8   | 6.4   | 13.4  | 21.2  | 23.4  | 24.7  | 22.7  | 16.9  | 10.1  | 5.5   | 4.2   |  |  |
|                                                                        | 2%          | WB     | 0.4   | 0.7   | 1.9   | 5.3   | 11.6  | 15.0  | 17.7  | 16.8  | 12.7  | 7.6   | 2.7   | 1.0   |  |  |
|                                                                        |             | MCDB   | 1.6   | 2.3   | 4.2   | 9.1   | 18.0  | 19.6  | 22.8  | 21.0  | 15.1  | 8.9   | 3.7   | 2.0   |  |  |
|                                                                        | 5%          | WB     | -1.4  | -0.9  | 0.7   | 3.9   | 9.5   | 13.5  | 16.5  | 15.4  | 11.6  | 6.1   | 1.0   | -0.2  |  |  |
|                                                                        |             | MCDB   | -0.7  | 0.0   | 2.7   | 7.1   | 14.9  | 18.3  | 20.9  | 19.2  | 13.7  | 7.5   | 1.5   | 0.4   |  |  |
|                                                                        | 10%         | WB     | -3.3  | -2.3  | -0.6  | 2.8   | 7.9   | 12.3  | 15.3  | 14.1  | 10.5  | 4.7   | 0.0   | -1.5  |  |  |
|                                                                        |             | MCDB   | -2.8  | -1.4  | 1.0   | 5.8   | 11.8  | 17.2  | 19.3  | 17.4  | 12.4  | 5.8   | 0.3   | -0.9  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 9.2   | 10.1  | 11.4  | 10.3  | 9.4   | 9.4   | 9.7   | 9.8   | 8.8   | 6.7   | 7.5   | 8.6   |  |  |
|                                                                        | 5% DB       | MCDBR  | 9.7   | 9.0   | 10.4  | 12.0  | 15.3  | 13.8  | 13.7  | 13.9  | 11.4  | 8.4   | 6.9   | 8.1   |  |  |
|                                                                        |             | MCWBR  | 8.9   | 7.8   | 8.2   | 7.5   | 8.3   | 6.4   | 6.5   | 7.6   | 7.5   | 6.5   | 6.1   | 7.5   |  |  |
|                                                                        | 5% WB       | MCDBR  | 9.0   | 8.3   | 9.9   | 11.0  | 13.7  | 11.4  | 11.4  | 12.1  | 9.6   | 7.6   | 6.4   | 7.9   |  |  |
|                                                                        |             | MCWBR  | 8.4   | 7.4   | 8.0   | 7.2   | 7.7   | 6.1   | 6.1   | 7.1   | 6.7   | 6.1   | 5.8   | 7.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.181 | 0.244 | 0.281 | 0.311 | 0.329 | 0.333 | 0.341 | 0.326 | 0.295 | 0.268 | 0.344 | N/A   |  |  |
|                                                                        | taud        |        | 1.932 | 2.132 | 2.213 | 2.258 | 2.451 | 2.498 | 2.487 | 2.511 | 2.574 | 2.377 | 2.793 | N/A   |  |  |
|                                                                        | Ebn at Noon |        | 184   | 594   | 771   | 845   | 871   | 876   | 855   | 831   | 781   | 582   | 223   | 0     |  |  |
|                                                                        | Edh at Noon |        | 12    | 49    | 79    | 99    | 92    | 91    | 89    | 78    | 57    | 39    | 15    | 0     |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.06  | 0.56  | 2.04  | 3.91  | 4.91  | 5.08  | 4.75  | 3.47  | 1.93  | 0.67  | 0.12  | 0.00  |  |  |
|                                                                        | RadStd      |        | 0.00  | 0.06  | 0.12  | 0.18  | 0.45  | 0.40  | 0.47  | 0.30  | 0.26  | 0.11  | 0.01  | 0.00  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (42 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page